

UAI301 Object-Oriented Programming
Lab Assignment 3
Topic: Classes, Objects, Constructors, and Methods

1. Create a class named Student. Create an object of the class and display a message: “Welcome to Object-Oriented Programming using Python”.
2. Create a class Student having the following attributes: Name, Branch, and Semester. Initialise the attributes using a constructor and display the details using a member function.
3. Create a class Employee with attributes: Employee ID, Employee Name, and Salary. Display the details of two employees.
4. Create a class Rectangle. The constructor should accept: Length and Breadth. Create a function to calculate the area.
5. Create a class Circle. Input the radius through the constructor and calculate the area of the circle. (Use $\pi = 3.14$)
6. Create a class BankAccount. The class should have: Account Number, Account Holder Name, and Balance. Create methods: Deposit, Withdraw, and Display Balance.
7. Create a class Book. Store: Book ID, Title, and Author. Create three objects and display their details.
8. Create a class Student. Input: Name, and Marks in three subjects. Create a method to calculate: Total Marks, Percentage.
9. Create a class Laptop with attributes: Brand, Processor, RAM and Price. Create a method to display all specifications.
10. Create a class LibraryBook. Store: Book Name, Author, and Number of Copies. Create methods: Issue Book (decrease copies by 1 if available), Return Book (increase copies by 1), and Display Details
11. Create a class Calculator with methods: add(), subtract(), multiply(), and divide(). The numbers should be provided through the constructor.
12. Create a Car class with attributes brand, model, year, and price. Use a constructor to initialize the attributes. Create methods to display the car details and update the price.
13. Create a MobilePhone class with attributes brand, model, storage, and battery_percentage. Use a constructor to initialize the details. Implement methods to display phone information, charge the battery, and check the battery status.
14. Create a Product class with attributes product_name, product_id, price, and quantity. Implement methods to calculate the total bill and display product details.
15. Create a BookStore class with attributes book_name, author, price, and number_of_copies. Implement methods to display book details, sell a book, and add new copies.
16. Create an Attendance class with attributes student_name, total_classes, and attended_classes. Implement methods to calculate the attendance percentage and display whether the student is eligible to appear in the examination.